

Robust Multi-robot Task Assignment in Surveillance

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Abstract:

A market based approach for the assignment of tasks in a multi-robot system is presented in this paper. The mobile robots are organized in groups, where tasks are locally assigned using an auction. The performance of the groups is evaluated with a measure similar to the concept of economic prices. The robots use this information at run-time, to switch to groups with higher prices. A theoretical analysis is presented and it is shown that under reasonably mild assumptions: (i) tasks are always assigned if there are idle robots, (ii) it is possible to detect the occurrence of failures which prevent the assignment of tasks and (iii) the approach is robust to the breakdown of any individual component. The range of possible choices for dividing the robots in groups is quantified with an entropy based measure. It is shown that the entropy value decreases if the components suffer severe failures or if the operating conditions deteriorate.

Keywords: mobile robot, task assignment, auction

1. INTRODUCTION

A approach for the assignment of tasks in multi-robot systems is presented, based on previous work by the authors, Gonçalves and Sequeira (2009). In this paper, is introduced a decentralized mechanism organizing the robots in groups. The result is an increase in the robustness of the task assignment system.

The target application considered is surveillance, where the mobile robots are employed to intercept intruders, block their exit pathways or attempt to take their mugshots. The purpose is to assist in the early prevention of damages and reduce the exposure to threats of human security forces.

The surveillance environments are typically dynamic and have an high degree of uncertainty. The behavior of the intruders is also not known beforehand and is difficult to model. Thus, some general design assumptions must be made. It is assumed that: (i) the down time of the communication network is small and (ii) estimates are known for the worst case values of the tasks arrival rates of and execution times. Furthermore, it is assumed that: (iii) a robot can execute only one task at a time, (iv) the execution cannot be preempted or interrupted and (iv) all tasks have the same priority.

The first three assumptions define the expected operating conditions for the proposed task assignment system. Although they may change abruptly, it is not feasible to develop a task assignment system to cover all possible contingencies. Nevertheless, it is shown in Section 4 that a feedback signal is available from the system, useful for diagnostic purposes. This is an important feature in surveillance applications, since it allows to infer directly from the system when it is not operating properly.

The last two assumptions may appear unreasonable for a surveillance application, because the execution of some tasks can be more urgent than that of others. However, no assumptions were made on the behavior of the intruders, the structure of the environment or the semantics of tasks. Without these additional assumptions, it is not possible to determine which tasks have a higher priority or those which can be safely interrupted by others. To better illustrate this point, suppose that an intruder was detected by a robot during a patrol route. Without further assumptions on the distribution of the intruders in the environment, it cannot be argued that tasks related to interception and identification have a higher priority than patrol tasks. It may well happen that more intruders exist along the route and will not be discovered if the patrol task is interrupted.

The remainder of this paper is as follows. A small review of the literature on related work is presented in Section 2. The system for task assignment is described in Sections 3 and 4. The design is inspired by concepts from economic theory, but it must be stressed that these concepts are only metaphors. The robots do not have preferences over the set of tasks nor do they compete, in sense of game theory, for their assignment. The approach is evaluated in a set of simulations, presented in Section 5.

2. RELATED WORK

The problem of assigning tasks to workers when the arrival order is known and no failures occur is in general NP-Hard Jungwattanakit et al. (2009). In some cases it is possible to solve this problem efficiently. For example, if all tasks arrive simultaneously, the assignment solution is computed efficiently with the Hungarian algorithm, Burkard (2002).

The approach proposed is inspired by economic markets, where an auction is used to assign tasks. The use of auctions to solve

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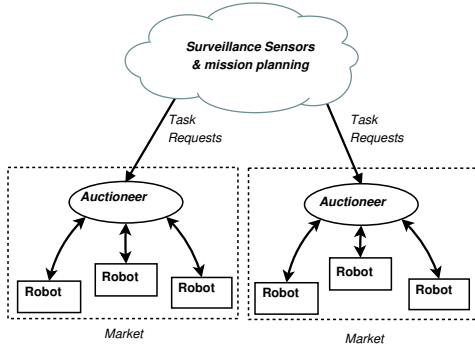


Fig. 1. Task assignment system

general assignment problems has been treated and analyzed in Brydon (2006); Nisam et al. (2007), while in Smith (1980) they were applied to distributed, collaborative problem solving. A survey on the use of auctions in mobile robotics can be found in Dias et al. (2006). The most common task is navigating along a set of way-points, which does not require a tight coordination between the robots. Nevertheless, auctions have also been applied in formation control problems, Brunet (2008). The auctions have been employed in different fields, but are often tailored to suit specific assignment problems, Nisam et al. (2007). For this reason, usually only possible the computational and communication complexities of auctions are compared Dias et al. (2006); Nisam et al. (2007). Also, only a few performance bounds and metrics are available which do not depend on the application Gerkey and Mataric (2003). In mobile robotic applications, the values of the bids submitted are often determined *a priori* by the system designer Dias et al. (2006). Since the tasks are often related to navigation in the environment, the bids are a combination of the travel time and distance Gerkey and Mataric (2003). However, the computation of bids is not trivial for the case of heterogeneous multi-robot system. An approach was proposed in Spaan et al. (2009), where the execution of tasks is modeled by a class of Markov Decision Problems (MDP). The bids are obtained directly from the value functions that are computed when solving the MDP. An advantage of this approach is that uncertainty in the knowledge of the state is naturally included in the bid values.

3. TASK ASSIGNMENT SYSTEM

The proposed approach is represented in Figure 1, where the arrows indicate communication links. The robots are divided in groups, denoted *markets*, and their initial composition is decided by the designer. The only constraint is that robots cannot simultaneously be members of more than one market.

A task is represented by τ and is identified with the concept of services in computer science, Specification (2006). Similarly to services, the manner in which tasks are executed is known only by the robots and does not depend of the tasks which where executed previously. The requests executing of tasks are received from the surveillance system sensors and the mission planning. This component and distributions of the detection events are not known and the requests can arrive at any instant and in any order. They sent to one of the *auctioneers*, either at random or based on predefined filters. For instance, each region of the environment is assigned to a market. The robots are then assigned the tasks by the auctioneers, through an auction. It is repeated periodically and the requests which were not assigned

in a round, accumulate to the following rounds until being assigned. The auctioneer is a critical point of failure in the market, and in case of a breakdown the tasks are no longer assigned. This is a common with auction, Dias et al. (2006); Nisam et al. (2007), although a decentralized auction was proposed by Brunet (2008). The assignment solution is computed with a greedy algorithm and does not have any optimality guarantees. Also, in heterogeneous multi-robot systems it can occur that some tasks are not assigned when there are sufficient robots available.

Since the approach is relies on multiple markets, it is robust to the individual failure of auctioneers. However, the quantity of tasks waiting to be assigned at each auctioneer can vary greatly. This occurs when an auctioneer receives much more tasks than the others or when there not enough robots in the market. The result will be an increase in the assignment delay of tasks in that market, Gonçalves and Sequeira (2009). The authors initially proposed for auctioneers to trade among them the tasks surplus. The solution is effective in balancing the workload of all auctioneers, but is still vulnerable to the individual breakdowns. Therefore, a new solution is proposed in which the robots are allowed to switch between markets. The idea is for robots to become members of markets where the auctioneers have high workloads. The robots can also leave their market if the respective auctioneer is not working properly.

3.1 Task Assignment in Markets

The procedure for the assignment of tasks in a market is described in full detail Gonçalves and Sequeira (2009). Due to the lack of space, it is only succinctly described here. The auctioneers periodically announce the tasks received and not yet assigned, to the robots in their market. They reply within a limited time frame, with their fitness values for each of the tasks announced. Before the next announcement, the auctioneer solves a mixed-integer assignment problem and informs the auction winning robots. The procedure is repeated periodically, with period Δ , until there no more tasks remaining.

Since the assignment is periodic, the tasks are always assigned with a delay which is always greater than Δ , see Gonçalves and Sequeira (2009). The delay cannot be made arbitrarily small because of the network transmission delays and limited bandwidth. Thus, the assignment delay is thus an indirect measure on the quality of service of the network.

It was shown in Gonçalves and Sequeira (2009), that the assignment delay for any task is less or equal than 2Δ if and only if

$$N \geq \Delta \sum_{\tau} a_{\tau} (1 + s_{\tau}) \quad (1)$$

where N is the number of robots in the market and $a_{\tau} \in \mathbb{R}_+$ is the maximum number of task arrivals per period Δ . The maximum execution of time of a task is $s_{\tau} \Delta \in \mathbb{Z}^+$. The upper bound is not tight because the number of robots is an integer variable. It is possible for small increases in a_{τ} and s_{τ} to occur without the corresponding increase in the assignment delay.

4. AUTONOMOUS MARKET SELECTION

In economic science, the prices of services and goods measure the efficiency of the rules that guide the interactions between

agents Gode and Sunder (1993); Nisam et al. (2007). The market rules are efficient if they can produce an equilibrium between the demand and supply, resulting in stationary, or quasi-stationary, prices Gode and Sunder (1993).

The same principle is applied to evaluate the performance of each of the markets. From the perspective of economic theory, the robots in a market are identified with the supply for the execution of tasks. The demand is then the number of task announced at the beginning of the auction round. The ratio between these two quantities is then the price of tasks.

Definition 1. (Market Task Price). The price of executing tasks of type τ in the market m at instant k is

$$p_m(k, \tau) = \frac{w_m(\tau, k-1)}{1 + y_m(\tau, k)}$$

where $w_m(\tau, k-1)$ is the number of tasks announced at the start of the auction round and $y_m(\tau, k)$ is the number of assigned tasks.

They are higher in markets with fewer robots or with more accumulated tasks, all other things being equal. The prices can be understood as resulting from the law of supply and demand, but it must remain clear that this is a metaphorical explanation. The robots do not have a bidding strategy to maximize their revenue. The auctioneers also do not attempt to maximize the price at which tasks are assigned.

The prices are determined by the interaction between the environment, the robots and the auctioneers and it was shown in Gonçalves and Sequeira (2009) that prices are always less than one if and only if the condition (1) is verified. Since the upper bound on the assignment delay is related to the operating conditions, the prices provide a feedback signal on the operation of each market. It was already available from the tasks assignment delays, but the market prices aggregate this information on a single measure that is more convenient to share with the rest of the system. In particular, the robots decide to switch markets based their price.

Definition 2. (Market Selection Procedure). The robots know to which market they belong and receive from all auctioneers, their local task prices. With a periodicity $\Theta > \Delta$, the idle robots not participating in an auction are allowed to switch markets. They randomly select between remaining in the current market or leaving for any with prices equal or greater than one.

The purpose of the procedure is thus for robots to switch to markets with higher task prices. In this manner, the supply of robots in those markets is increased. The proposed procedure is robust to the breakdown of some components, either auctioneers or robots. The robots that suffer a breakdown do not prevent the others from switching markets since there is no explicit coordination among them. The same is valid for the auctioneers, those that suffer a breakdown do not announce their prices and thus no robots will switch to their markets.

4.1 Market Switching Behavior

The switching procedure has no explicit stopping condition or coordination among the robots. Since prices are kept low in markets with a sufficient number of robots, the departure of some may cause prices to increase locally. Therefore, it is important to determine if the switching behavior will eventually stop or instead continue indefinitely.

It is assumed that the auctioneers announce their prices simultaneously and that all robots also select their next market simultaneously, albeit independently. Furthermore, the selection period Θ is assumed not to be a multiple of Δ . The task assignment system can operate without these assumptions, but the market selection dynamics become much harder to model and analyze. The assumptions require that the synchronization between the clocks of all robots and auctioneers. This is not trivial to accomplish in practice, although some solutions exist, see for instance Verissimo and Rodrigues (2001); Schenato and Gamba (2007).

Let the time instants at which markets are selected be indexed by the discrete variable d . The number of mobile robots in market m at instant d is $x_m(d)$. The set of markets is M , with cardinality $\#M$. The arrival rates of tasks at each market are represented by the elements of matrix $A = [a_\tau^m]$. Their maximum execution times is assumed to be independent of the market and represented by the elements of vector $s = [s_1, s_2, \dots]'$. These two variables represent the configuration of the environment in which the system is supposed to operate.

The number of robots in a market at instant d is a function of the number of robots in all of the markets at the previous instant and the environment configuration. Thus, the dynamic behavior of the selection procedure can be modeled by the Markov chain

$$x(d) = h(x(d-1), A, s) \quad (2)$$

where x is the state vector, $x = [x_1, \dots, x_{\#M}]'$ and $h(\cdot)$ is the transition function. The configuration of the environment is assumed to be quasi-stationary, with abrupt changes occurring when components breakdown or a large number of intruders are detected. The Markov chain is homogeneous because the probability of a robot switching to a different market does not vary with time. The time variable is made explicit because we are interested in characterizing the trajectories of the chain, not the states distributions.

The condition (1) is fundamental to the behavior of task prices and the market switching. Thus, it is convenient to represent it in a compact form

$$\phi(x_m(d), A, s) = \begin{cases} 0, & \text{if } x_m(d) \geq \Delta \sum_\tau (1 + s_\tau) a_\tau^m \\ 1, & \text{otherwise} \end{cases} \quad (3)$$

Proposition 1. (Markov Chain States). Assume that for a static configuration of the environment, there are enough robots such that

$$\exists x' \in X \Rightarrow \forall m \in M, \phi(x'_m, A, s) = 0 \quad (4)$$

Then any state of the Markov chain is either absorbent or transient.

Proof The proof of this statement is done in two steps. First it is shown that there are absorbing states in the Markov chain if there are sufficient robots. Then it is shown that with strict positive probability, any absorbing state can be reached from any other state in a finite time.

First step

Let the probability at instant d of market m having n robots which can switch to other markets be $\psi(m, n, d)$. Assume that

$\psi(m, n, d) > 0$ for all $n = 0, \dots, x_m(d)$ and $\psi(m, n, d) = 0$ only if $n > x_m(d)$. This is reasonable since no assumptions were made on the distributions of the arrival rates or the service times, although the values $\psi(m, n, d)$ can be very small. The probability that no robots leave market m at instant d is

$$Prob(d, m, 0) = \sum_{n=0}^{x_m(d)} \psi(m, n, d) \cdot \left[1 - \frac{\sum_{l \in M \setminus \{m\}} \phi(x_l(d-1), A, s)}{1 + \sum_{l \in M \setminus \{m\}} \phi(x'_l(d-1), A, s)} \right]^n$$

Therefore, the probability that no robots switch markets is $Prob(x(d), 0) = \prod_{m \in M} Prob(d, m, 0)$. If the task demand is satisfied in all markets at the previous instant, then

$$\begin{aligned} \forall m \in M : \phi(x_m(d-1), A, s) &= 0 \\ \Rightarrow \sum_{l \in M \setminus \{m\}} \phi(x_l(d-1), A, s) &= 0 \\ \Rightarrow Prob(d, m, 0) = \sum_{n=0}^{x_m(d)} \psi(m, n, d) &= 1 \\ \Rightarrow Prob(x(d), 0) &= 1 \end{aligned}$$

Since no robots switch to a different market at time d and assuming the environment configuration to be static, then $x(d') = x(d)$ for all $d' \geq d$. This is the definition of an absorbing state of the Markov chain.

Second Step

Consider the state trajectory $x(d)$, with $x(0) = x^0$ and $x(d^1) = x^1$ for all $d \geq d^1$. At each selection instant, exactly one robot switches between markets until x^1 is reached. The required minimum number of switching instants d^1 is

$$\max_{m \in M} |x_m^1 - x_m^0| \leq d^1 \leq \sum_{m \in M} |x_m^1 - x_m^0|$$

and thus the absorbent state x^1 can be reached in a finite number of steps. The probability of only one robot switching between markets at each state $x(d)$ is

$$Prob(x(d), x(d+1), 1) = \sum_{m \in M} Prob(d, m, 1) \cdot \prod_{m' \in M \setminus \{m\}} Prob(d, m', 0)$$

where $Prob(d, m, 1)$ is the probability of only one robot in market m to leave for another market. It has a binomial distribution

$$\begin{aligned} Prob(d, m, 1) &= \sum_{n=1}^{x_m(d)} \psi(m, n, d) \binom{n}{1} Prob(m, d, 1^*) \cdot [1 - Prob(m, d, 1^*)]^{n-1} \\ &= Prob(m, d, 1^*) \cdot \sum_{n=1}^{x_m(d)} n \cdot \psi(m, n, d) \cdot [1 - Prob(m, d, 1^*)]^{n-1} \end{aligned}$$

where the term $Prob(m, d, 1^*)$ is the probability a robot in m leaving for another:

$$Prob(m, d, 1^*) = \frac{\sum_{l \in M \setminus \{m\}} \phi(x_l(d-1), A, s)}{1 + \sum_{l \in M \setminus \{m\}} \phi(x_l(d-1), A, s)}$$

The probability $Prob(x(d), x(d+1), 1)$ is always strictly greater than zero because $Prob(d, m, 0) > 0$ for any state $x(d)$ and also $Prob(d, m, 1) > 0$ as long as there is a market in state $x(d)$ such that $\phi(x_l(d-1), A, s) = 1$. That is, $Prob(d, m, 1) > 0$ if the state $x(d)$ is not absorbent. Since the absorbent state x^1 can be reached in a finite number of steps with a strictly positive probability, the chain will never return to state x^0 with positive probability. This is the definition of a transient state in a Markov chain, Cassandras and Lafortune (2009), and therefore all non-absorbent states are transient.

□

A corollary of this proposition is that the switching procedure does not stop if the tasks demand cannot be satisfied. The reason is that in this case all of the states in the Markov chain are recurrent. The proof is identical to the second step of the proof for Proposition 1 and is not repeated here.

The dynamic behavior of the switching procedure is described qualitatively by Proposition 1. Since an absorbing state can always be reached from any other state, the switching procedure will stop eventually. This is important since the procedure has no explicit stopping condition and the robots do not coordinate their individual decisions. In practice however, it can take a long time for the Markov chain to reach an absorbent state. This is not an important issue since the task prices always inform the human supervisors that the system is not operating properly. If the task prices persist for too long, the supervisors can execute appropriate contingency measures.

4.2 Organization Entropy

A difficulty with the approach proposed is how to properly organize the robots among the markets. The initial solution is computed from (1), if the configuration of the environment is known. However, in general the solution is not unique and the operating conditions can change abruptly. Thus, it is important to use an initial solution that is robust to small variations of the environment configuration. The uncertainty on how to initially organize the robots is quantified by a measure of entropy, denoted the *organization entropy*. Intuitively, it is identified with the number of absorbing states of Markov chain of (2). The higher their number, the larger the range of possible choices for dividing the robots among the markets.

Suppose that the designer has available a fixed number of robots, N , and has decided on the number of markets, $\#M$. Let $\bar{X}(A, s, N, \#M) \subset X$ be the non-empty subset of absorbent states of the Markov chain (2). Since the chain will eventually reach an absorbent state, let $P(\bar{x})$ be the probability of reaching state $\bar{x} \in \bar{X}(A, s, N, \#M)$ from any other non-absorbent state. The concept of entropy was generalized by Rényi (1960) to a family of parametrized functions in the form

$$H_\alpha(P_1, P_2, \dots, P_r) = \frac{1}{1 - \alpha} \log \left(\sum_{k=1}^r P_k^\alpha \right) \quad (5)$$

where P_i is the probability for the occurrence of the i -th discrete event and α is a real valued parameter, $\alpha > 0$ and $\alpha \neq 1$. The Shannon entropy, for instance, is obtained as the limit

case when α tends to one. The Rényi entropy of the subset of absorbent states is

$$H_\alpha(\bar{X}(A, s, N, \#M)) = \frac{1}{1-\alpha} \log \left(\sum_{\bar{x} \in \bar{X}(A, s, N, \#M)} P^\alpha(\bar{x}) \right) \quad (6)$$

Since the organization entropy is identified with the number absorbent states, it is the limit case of $H_\alpha(\bar{X}(A, s, N, \#M))$ where α tends to zero

$$\begin{aligned} H_0(\bar{X}(A, s, N, \#M)) &= \lim_{\alpha \rightarrow 0} H_\alpha(\bar{X}(A, s, N, \#M)) \\ &= \log(\#\bar{X}(A, s, N, \#M)) \end{aligned} \quad (7)$$

The organization entropy is thus proportional to the number of absorbent states. It has a complex dependency from multiple factors: the number of robots, the number of markets and the environment conditions. Nevertheless, it is clear that the occurrence of breakdowns in the robots or changes in the environment conditions lead to changes in the organizational entropy. The best initial organization of the robots does not necessarily maximize the organizational entropy. Instead it is the one in which entropy increases if the robots suffer failures or the operating conditions change. In this manner, the number of absorbent states increases when the system is not operating properly.

5. SIMULATION RESULTS

The proposed approach is applied in simulation to the urban environment of t URUS project Sanfeliu and Andrade-Cetto (2006). It consists of a square in the UPC campus at Barcelona, Spain, with approximately 100 meters in each side. The P3-AT mobile robots are available, with the maximum linear velocity limited to $0.3m/s$ for safety and to extend the battery life. The assignment period is $\Delta = 3s$ and the robots must: i) visit a location, ii) recharge/swap batteries, iii) patrol the perimeter and patrol along each of the diagonals. The arrival rates and execution times of each task are in Table 1, based on those previously used in Gonçalves and Sequeira (2009). The arrival rate unit is the number of arrivals per assignment period. For instance, task $\tau = 1$ has a maximum arrival rate of 0.022 per assignment period or approximately 2 arrivals per hour.

Task τ	Arrival Rate, a_τ	Execution Time, s_τ [secs]
1	$0.022/\Delta$	180Δ
2	$0.005/\Delta$	258Δ
3	$0.0025/\Delta$	445Δ
4	$0.005/\Delta$	158Δ
5	$0.005/\Delta$	158Δ

Table 1: Task contracts parameters

Suppose that all robots are grouped in a single market. From (1), a minimum of eight robots are needed. The operation of the system was simulated for the period of a approximately one day, with the task requests generated at the maximal rate. The robots suffer breakdown events, exponentially distributed, which disable them permanently. The price of all tasks and the number of available robots are plotted in Figure 2. It is visible that when the number of robots is less than eight, the task prices

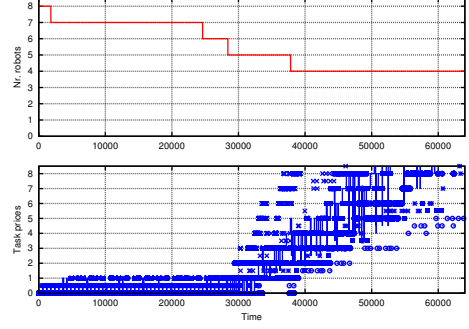


Fig. 2. The prices of each task in the single market

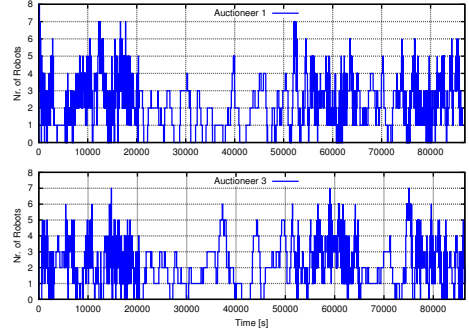


Fig. 3. Robots switching in two markets

are increased. As expected, the task prices provide a feedback signal on the operation of the architecture.

Consider now that the campus is divided in five regions, each with an auctioneer $\#M = 5$. The task requests received by the system are uniformly divided at random among the auctioneers. The arrival rates at each market are now $1/5$ of the values in Table 1. At least two robots are needed at each market, but the designer has only eight robots. The system was simulated for a period of one day and the number of robots in two of the markets are plotted in Figure 3. The switching does not stop because there are no enough robots to produce absorbent states. However, it was verified that the average number of robots in each market is, respectively, (1.96, 2.07, 1.35, 0.5), close to the required minimum. The fifth market as the lowest average because the distribution of tasks is only approximately uniform.

The organization entropy is a function of the environment configuration, the number of robots and auctioneers. It is plotted in Figures 4 and 5, as a function of these variables. The curves are not smooth because the number of robots and auctioneers are integer variables. The number of states grows rapidly with the number of robots and auctioneers, which also limits the number of points in the plots. The organization entropy is proportional to the number of robots, Figure 4, because if some robots suffer a breakdown, there are less options available for organizing the system. Similarly if more robots are added to the system, the number of choices increase. The curve for the variation of the entropy with the number of auctioneers, Figure 5, suggests that a low ratio of robots to markets should be used. In general, the failure of an auctioneer increases the number of robots available to other markets. However, this also increases the arrival rates of tasks at other markets. There is thus a trade-off, which can be settled with plots similar to Figure 5.

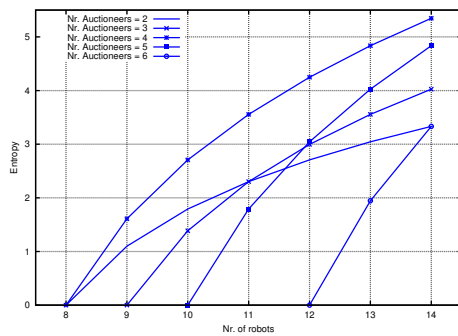


Fig. 4. Variation of the organization entropy with the total number of robots

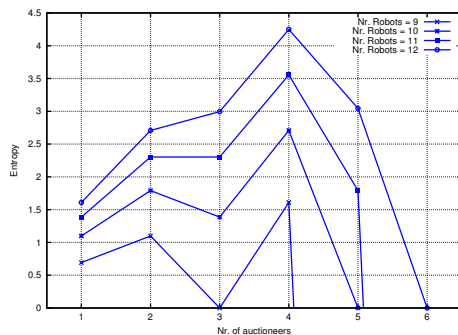


Fig. 5. Variation of the organization entropy with the number of auctioneers

6. CONCLUSION

A failure tolerant approach was presented for the assignment of tasks in multi-robot systems. A measure, denoted task price, was proposed for the automatic detection of failures that prevent the assignment of tasks in the markets. Although it does not identify the cause of the failures, it provides a feedback signal to the human supervisors and the mobile robots. The latter use the prices information to switch to the markets where tasks demand is not satisfied. The approach presented is robust to the failures of components, either robots or auctioneers. The auctioneers that suffer a breakdown do not prevent the local robots from switching to other markets. The break down of a robot also does not prevent the others from being assigned tasks or change markets.

The procedure for switching between markets is very simple and does not employ any explicit coordination mechanism or stopping condition. The switching behavior was modeled as a Markov chain and it was shown that it stops when the robots become organized such that prices in all markets are smaller than one. In the simulation of a chain without absorbent states, the switching procedure was observed to promote an average distribution for the robots close to the desired one. Although this is an empirical result, the procedure does aim at uniformly distributing the robots among all the markets with high prices.

A measure of entropy was proposed for evaluating the quality of the organization. The entropy curves provide a means for designers to select the number of auctioneers and robots to use, without requiring additional information from the environment. The curves agree also with what is expected to occur in practice. Thus, it is believed that the design and operation of the proposed system are supported on realistic assumptions.

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